



POLICY AND PROCEDURE

Medication: Clozapine Prescribing and Monitoring

Scope (Staff):	North Metropolitan Health Service Mental Health (NMHS MH) Staff: Clinical
Scope (Area):	NMHS MH All Areas

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1. Aim

This document provides NMHS MH clinical staff with the requirements for prescribing and management of clozapine to ensure safe and effective treatment for patients of the service.

2. Background

Clozapine is indicated for the treatment of schizophrenia where the patient is non-responsive to or intolerant of two other antipsychotics:

- Non-response is defined as lack of satisfactory clinical improvement despite the use of adequate doses of antipsychotic medications prescribed for a duration expected to achieve antipsychotic effect.
- Intolerance is defined as the inability to achieve adequate benefit with other antipsychotics because of severe and untreatable adverse reactions.

Clozapine is a High-Risk Medication as defined by the [WA High Risk Medication Policy](#). The significant risks are:

- Agranulocytosis
- Myocarditis
- Cardiomyopathy
- Clozapine Induced Gastrointestinal Hypomotility (CIGH)

A national patient monitoring system is in place to support the monitoring for the risk of agranulocytosis. The risk of myocarditis/cardiomyopathy is managed by clinical monitoring. The risk of CIGH is managed by clinical monitoring, early and assertive treatment and consideration of prophylactic aperients.

3. Risk

Patients may suffer adverse outcomes related to clozapine if it is not prescribed and managed in accordance with this policy.

4. Definitions

TPR: Temperature, pulse, respiration

MSE: Mental State Examination

ULN: Upper Limit of Normal

5. Principles

- All components of the clozapine patient monitoring system will be complied with. WA Health uses the ClopineCentral™ monitoring system in accordance with the current contract for purchase of pharmaceuticals.
- Patients, prescribers, pharmacists and the centre (hospitals, clinics, community pharmacies) must be registered with ClopineCentral™

- Patients must be provided with written information on clozapine and advised about the monitoring requirements and the provisions of the patient monitoring system.
- Clozapine Initiation and Titration Chart PMR60G must be used for all patients commencing or re-titrating clozapine.
- Any current psychotropic medication, particularly antipsychotic medication must be reviewed and rationalised when commencing clozapine and during titration. Antipsychotic monotherapy is desirable where possible.
- Concomitant use of other medications with the potential to cause agranulocytosis is discouraged and increased vigilance must be maintained where this is necessary.
- Concomitant use of other medications with the potential to cause constipation should be minimised where possible. Where these medicines are used, prophylactic aperients should be considered.
- A doctor should review the patient at least once a week for the first 18 weeks. Thereafter they should be reviewed every 4 weeks by a qualified clinician to assess mental and physical state.

6. Pre-commencement checklist

- Consider PBS Eligibility and continuity of supply for clozapine treatment. For non-PBS indications prescribers must follow the Non-Formulary process.
- Patient/guardian gives informed consent, or second opinion obtained (if applicable), and documented in the patient's Integrated Progress Notes.
- Check if the patient has a history/family history of:
 - drug-induced neutropenia or bone marrow disorders, or any other factors that might increase the risk of neutropenia or agranulocytosis on clozapine
 - cardiovascular disease that could increase the risk of cardiac-related adverse effects while on clozapine e.g. hypertension
 - diabetes mellitus, dyslipidaemia, or other metabolic disorders
 - epilepsy
 - history of thromboembolism
- Review and quantify smoking status (tobacco and cannabis use)
- Check pregnancy and breast feeding status
- Patient/Authorised Representative to sign the Clopine® Monitoring System Privacy Statement and file in the patient integrated progress notes.
- Treatment explained to the patient/guardian/carer with written and verbal medicines information and this documented in the patient's Integrated Progress Notes. For written information refer to [Choice and Medication](#).

6.1 Mandatory blood monitoring and registration

Patient registration requirements:

- Patient initials
- Date of birth
- Sex

-
- Blood group

Patients must be registered as active with the monitoring system and a registration number documented in the patient integrated progress notes and on their Clozapine Initiation and Titration Chart (PMR60G) prior to administration of clozapine.

Baseline blood tests (see **6.2**) must be performed prior to commencing clozapine with a specific requirement that white blood count (WBC), neutrophil count (NC), CRP and troponin are performed within 10 days of commencing clozapine. WBC and NC blood test monitoring continues weekly for the first 18 weeks and then monthly thereafter.

6.2 Baseline monitoring

The following tests must be completed prior to initiating treatment, or when recommencing after therapy interruption greater than three months (also refer to [Section 12](#) – Therapy Interruption):

- Physical examination (including an abdominal examination). Refer to [NMHS MH Physical Healthcare Management Policy](#).
- Blood Group
- Blood tests (required within 10 days prior to commencing clozapine): Full Blood Count (FBC), Troponin, C-Reactive Protein (CRP), and pregnancy test (if applicable).
- Blood tests (required within 6 months prior to commencing clozapine): Urea & Electrolytes (U&Es), Liver Function Tests (LFTs), fasting blood glucose level (BGL), glycated haemoglobin (HbA1c), lipid profile,
- Electrocardiogram (ECG) (within 10 days prior to commencing clozapine)
- Echocardiogram (ECHO). Within 12 months prior to commencing clozapine. If patient has relevant medical co-morbidities and/or significant substance use consider performing closer to commencement. Where a pre-treatment ECHO cannot be performed, it should be obtained as soon as possible after commencement.
- Screen for and treat any pre-existing constipation (within 10 days prior to commencing clozapine). Use the Rome IV screening criteria (see [Section 9.3](#)). Refer patients to a dietitian for review. Document baseline bowel habits.

Baseline investigations are to be documented on the Clozapine Initiation and Titration Chart (PMR60G).

7. Clozapine Commencement

Clozapine therapy may be commenced in the hospital or community setting.

Admission to hospital is only necessary where the risk to the patient requires close monitoring provided by a ward environment.

7.1 Suitability for community commencement

- Adherent with oral medication and monitoring requirements.
- Understands the need for regular physical monitoring and blood tests.



- Understands the possible adverse effects and what to do about them (particularly the rare but serious ones).
- Readily contactable (e.g. in the event of a result that needs follow-up)
- Can be reviewed daily during the early titration phase.
- Can collect medication every week or have medication delivered to their home.
- Will be likely to seek assistance out-of-hours if they experience potentially serious adverse effects (e.g. indicators of myocarditis or infection such as fever, malaise, chest pain).

In any setting, the decision to commence clozapine should be made in collaboration with the patient and any significant stakeholders (guardian, family, carer, accommodation service, GP etc).

7.2 Contraindications to community commencement

It is potentially unsafe to commence clozapine in the community for patients who have a history of:

- seizures;
- significant cardiac disease;
- unstable diabetes;
- paralytic ileus;
- blood dyscrasias;
- neuroleptic malignant syndrome or other disorder that increases the risk of serious adverse effects;
- previous severe side effects on titration of clozapine or other antipsychotics;
- unreliable or chaotic lifestyle that may affect adherence to the medication or the monitoring regimen;
- significant current abuse of alcohol or other drugs likely to increase the risk of adverse effects.

8. Dosing

The Clozapine Initiation and Titration Chart PMR60G must be used for the first 28 days of clozapine commencement in both inpatient and community settings.

Commence clozapine at a low dose and increase gradually to minimise side-effects. A suggested dose titration schedule is included on the PMR60G suitable for inpatients. Refer to **Appendix 1** for a slower titration schedule for community patients.

Once initial dose titration or retitration after an interruption in therapy is complete, maintenance therapeutic doses are usually between 200 to 450 mg/day (usual maximum of 600mg/day). A few patients may require larger doses to obtain maximum therapeutic benefit. Dose increments should not exceed 25-50mg per increase per week.

Existing antipsychotic therapy (including long-acting antipsychotic injections) should be rationalised upon initiation and during the titration phase. It is acknowledged that clozapine monotherapy is best practice.



8.1 Inpatients

The Clozapine Initiation and Titration Chart PMR60G has a suggested dosing regime, but a slower titration may reduce the incidence of adverse effects.

8.2 Community Patients

Commencement of clozapine in the community requires a slower and flexible titration schedule compared to the inpatient titration regimen. Do not follow the suggested dosing regime printed on the Clozapine Initiation and Titration Chart PMR60G. Refer to [Appendix 1](#). Community Commencement Schedule for Prescribing and Monitoring for suggested dosing titration.

The first dose is given in the morning at the CMHC and the patient is monitored for at least 3 hours. If the dose is well tolerated after 3 hours, the patient can go home. Clozapine can cause drowsiness and if affected the person should not drive a motor vehicle or operate machinery afterwards.

Blister packs should be used for the titration period to assist the patient in managing the frequent dose changes. The dispensing pharmacy must be able to adjust the blister pack at short notice. Ongoing use of blister packs may be useful, to aid adherence.

9. Monitoring at commencement (first 4 weeks)

Monitoring for inpatients and community patients should be documented on the Clozapine Initiation and Titration Chart PMR60G for the first 4 weeks. Vital sign monitoring must be documented on both the PMR60G and PMR38 as specified in section 9.1. For ongoing monitoring refer to [Appendix 2](#).

9.1 Vital Signs Observation Monitoring

There are specific observation monitoring requirements when commencing therapy as clozapine can cause marked postural hypotension. Initial observation monitoring is partly aimed at detecting and managing this.

Inpatient Setting

7-day initial observation regime on the front of the Clozapine Initiation and Titration chart PMR60G must be completed.

- TPR and postural blood pressure (BP) is to be taken prior to the first dose and then hourly for the first six hours. Consider timing of dose initiation so hourly observation do not continue into bedtime.
- On Days 2 - 7, TPR and postural BP must be taken twice daily.
- Observations in the first 7 days must be documented on both the Clozapine Initiation and Titration Chart PMR60G and the Adult Observation and Response Chart (AORC) PMR38.
- Observations are to be taken at a minimum daily from Day 8 onwards whilst in hospital and observations documented on the AORC PMR38.



The front of the Clozapine Initiation and Titration Chart PMR60G documents the escalation requirements where urgent medical team notification for review is required.

Community Setting

The observation monitoring requirements are different from inpatient monitoring.

- TPR and postural BP are checked prior to the first dose of clozapine and then checked hourly for the next 3 hours.
- All initiations should take place early in the week so that adequate staffing and monitoring are assured.
- Required monitoring will be done at the CMHC

Refer to [Appendix 1](#) for the full list of daily observation monitoring requirements.

9.2 Monitoring for agranulocytosis and neutropenia

WBC & NC must be monitored weekly for the first 18 weeks of treatment and 4 weekly thereafter for the duration of treatment.

More frequent monitoring is required as per the monitoring protocol if WBC and/or NC declines or if the patient exhibits clinical symptoms of infection (e.g. fever, sore throat).

Clozapine Blood Results Monitoring System		Recommended Action
Green Range	WBC greater than $3.5 \times 10^9/L$ AND Neutrophils greater than $2.0 \times 10^9/L$	Continue clozapine therapy
Amber Range	WBC $3.0 - 3.5 \times 10^9/L$ AND/OR Neutrophils $1.5 - 2.0 \times 10^9/L$	Continue clozapine therapy with twice-weekly blood tests until return to "green" range
Red Range	WBC less than $3.0 \times 10^9/L$ AND/OR Neutrophils less than $1.5 \times 10^9/L$	Stop clozapine therapy immediately. Contact haematologist and Clozapine Monitoring Centre

9.2.1 Benign Ethnic Neutropenia

Benign ethnic neutropenia (BEN) is a rare, hereditary condition in which the neutrophil count is relatively low and may frequently fall into the lower limit of normal.

The presence of benign ethnic neutropenia should not prevent treatment with clozapine. Haematologist consultation prior to initiating clozapine is recommended as this may lead to an adjustment of the white blood cell count green, amber and red ranges. Any adjustments must be approved by the clozapine patient monitoring system ClopineCentral™ and will be logged as a note on the patient's profile.

9.3 Monitoring for Clozapine Induced Gastrointestinal Hypomotility (CIGH)

CIGH is increasingly recognised as a highly prevalent and potentially fatal adverse effect of clozapine. Active monitoring of patients, including direct questioning, is essential.

All hospital inpatients require a Bowel Chart to be used and patients are to be asked about their

bowel habits using the Rome IV criteria (see below):

- Baseline
- Daily for the first 4 weeks
- A minimum of weekly thereafter for the duration of their hospital stay.

For patients in the community, bowel habits must also be monitored:

- At baseline and then asked and documented at every review (weekly as a minimum requirement) for the first 18 weeks of clozapine treatment, and then four weekly thereafter for the duration of treatment.

Some patients may not report even life-threatening constipation. For this reason prophylactic laxatives should be considered for patients prescribed clozapine. First line options include stimulant and osmotic laxatives. Bulk-forming laxatives like Metamucil® should be avoided.

Extra vigilance is required when other medications that cause constipation are co-prescribed. Smoking cessation and infective illness can also increase the risk of severe constipation due to elevated clozapine levels.

Monitoring should be based on the following criteria:

Rome IV assessment criteria for functional constipation

Must include at least two of the following:

- straining during at least 25% of defecations
- lumpy or hard stools in at least 25% of defecations
- sensation of incomplete evacuation for at least 25% of defecations
- sensation of anorectal obstruction/blockage for at least 25% of defecations
- manual manoeuvres to facilitate defecation for at least 25% of defecations (e.g. digital evacuation, support of the pelvic floor)
- fewer than three spontaneous bowel movements per week

Assertive treatment of any emerging constipation is necessary including a review of other medication that may be contributing (e.g. anticholinergic, opioid medications).

Signs and symptoms that warrant immediate medical attention include moderate to severe abdominal pain, abdominal distension, vomiting, overflow diarrhoea and signs of sepsis.

9.4 Monitoring for myocarditis

Myocarditis is a hypersensitivity response to clozapine and can be potentially fatal. It most commonly occurs in the first four weeks (median 3 weeks). The patient must be monitored for



clinical features of myocarditis/infective illness throughout treatment, and the first four weeks of clozapine commencement is to be monitored and managed as per the following table:

Baseline	Pulse, BP, temp, RR CRP Troponin ECG Echocardiogram
Daily for inpatients and as per Appendix 1 for community patients	Pulse, BP, temp, RR Ask about features of myocarditis/infective illness (e.g. fever, cough, SOB,)
Days 7, 14, 21, and 28	CRP Troponin ECG if possible
If clinical features of myocarditis/infective illness develop	Daily CRP and troponin until condition resolves or an alternative cause is identified
If CRP 50-100mg/L or there is a mild elevation in troponin (≤ 2 ULN)	Continue clozapine with daily troponin, CRP and ECG until abnormality resolves. Do not increase clozapine during this time.
If CRP >100mg/L or Troponin > 2X ULN clozapine	Stop clozapine; repeat ECHO and request assessment by physician or cardiologist.

9.5 Monitoring for cardiomyopathy

- Cardiomyopathy occurs later in treatment than myocarditis (median of 9 months) but may occur at any time.
- Clinical presentation varies somewhat, and it is often asymptomatic in the early stages so any reported symptoms of palpitations, decreased exercise capacity, new onset fatigue, breathing difficulties, chest pain, syncope and sweating should be closely investigated. An ECHO is to be requested and cardiac biomarker can be checked.
- Persistent tachycardia can predispose patients to cardiomyopathy.

10. Ongoing Monitoring (>4 weeks)

Following on from the Clozapine Initiation and Titration Chart PMR60G.

Refer to [Appendix 2](#) for information regarding ongoing monitoring for inpatients and community patients.

11. Escalation Requirements

If at any time the patient experiences the following, this must be reported to the prescriber:

- **temperature rises above 38 °C** (this is common and is not a good reason, on its own, for stopping clozapine, this may also be linked to myocarditis)
- **pulse is >100 bpm** (also common and not, on its own a reason for stopping, but may sometimes be linked to myocarditis)
- **respiratory rate <10 or > 22 breaths/ minute**
- **postural drop of >30 mmHg**

- **patient is clearly over-sedated**
- **any signs of constipation**
- **flu-like symptoms** (malaise, fatigue, etc.)
- **chest pain, dyspnoea, tachypnoea**
- **any other adverse effect that is intolerable**
- **patient has hypersalivation** (utilise management strategies, potentially including medications)
- **changes in smoking habit**

12. Therapy Interruption

If patients decline blood monitoring or therapy is disrupted, the clozapine monitoring protocol must be followed. Clozapine treatment must not be continued without haematological monitoring.

Recommencing clozapine after interruption		
Time since last clozapine dose	Dose	Blood test monitoring
< 48 hours	Re-start at previous dose – no re-titration required.	No changes to monitoring
≥ 48 hours and ≤ 72 hours	Obtain psychiatric review prior to recommencing. Re-start dose at 12.5mg once or twice daily on the first day. If this dose is tolerated, it may be feasible to titrate the dose to the therapeutic level more quickly than is recommended for initial treatment.	No changes to monitoring
>72 hours and ≤ 4 weeks	Obtain psychiatric review prior to recommencing. Re-start dose at 12.5mg once or twice a day on the first day. If this dose is tolerated, it may be feasible to titrate the dose to the therapeutic level more quickly than is recommended for initial treatment.	Weekly monitoring schedule – Monitor weekly for 6 weeks or for as long as necessary to achieve a total of 18 weeks of weekly monitoring. Monthly monitoring schedule – monitor weekly for 6 weeks then continue with monthly monitoring if no problems detected.
>4 weeks and < 3 months	As per above	Weekly monitoring for 18 weeks
≥ 3 months	Recommence as for a new patient. Patient must be re-registered with the clozapine patient monitoring system	

13. Discontinuing Clozapine

Discontinuing clozapine carries a risk of rebound psychosis. A slow withdrawal reduces this risk. The patient's mental state and symptoms related to cholinergic rebound should be monitored.

Post therapy blood testing

If clozapine is ceased the post monitoring protocol must be followed.

- Patients on weekly blood tests, require weekly tests (WBC & NC) for 4 weeks after discontinuing therapy.
- Patients on monthly blood tests, require one further blood test one month after discontinuing therapy.

14. Rechallenge

If clozapine is ceased as a result of neutropenia or agranulocytosis, rechallenge requires haematologist approval as per the monitoring protocol.

Where patients have been trialled on clozapine and ceased for non-haematological reasons rechallenge is at the discretion of the treating psychiatrist with appropriate input from relevant specialties (for example – cardiologist post-myocarditis or gastroenterologist post-CIGH). Reasons for clozapine cessation and why a retrial is considered appropriate must be documented in the patient's medical records.

15. Transition of Care

Patients on clozapine who are discharged from an inpatient service must be discharged to the care of a psychiatrist engaged with a registered clozapine centre for ongoing prescriptions and monitoring.

Suitable patients on clozapine who are clients of a community mental health clinic may be transitioned to a shared care arrangement with a GP who is registered with a clozapine centre under the supervision of a clinic psychiatrist [NMHS MH Medication: Clozapine Management by General Practitioners](#).

16. Compliance and Evaluation

- All patients will have baseline monitoring parameters and registration completed prior to commencing treatment with clozapine.
- Zero reports on Datix CIMS involving non-adherence to this clozapine policy.

Related internal policies, procedures and guidelines

[NMHS MH Medication: Management of High Risk Medication Policy](#)
[NMHS MH Medication: Clozapine Management by General Practitioners](#)
[NMHS MH Medication: Clozapine Coordinator – Clozapine Clinic](#)

References

[ClopineCentral™ Protocol](#)
[Guidelines for the use of the WA Clozapine Initiation and Titration Chart](#)
[Clozapine Monitoring Form A](#)

[Clozapine Monitoring Form B](#)

Useful resources

[eTG: Psychotropic Version 8](#)

[Maudsley Prescribing Guidelines 13th Edition](#)

[Drug Bulletin. Clozapine Initiation Chart – A new medication chart for clozapine commencement. Graylands Pharmacy Department.](#)

[Medicines Safety Update. No. 1; 2011. Clozapine and severe constipation.](#)

[Choice and Medication](#)

[Drug Bulletin: Psychotropic drugs – the effect of Smoking and Caffeine](#)

[Drug Bulletin: Interpretation of Clozapine Levels](#)

[Drug Bulletin: Clozapine-induced gastrointestinal hypomotility Vol 26 No.3 Nov 2019](#)

[Drug Bulletin: Treatment of Clozapine-Induced Hypersalivation Vol 26 No. 1 Feb 2019](#)

Ronaldson KJ, Fitzgerald PB, Taylor AJ, Topliss DJ, McNeil JJ. A new monitoring protocol for clozapine-induced myocarditis based on an analysis of 75 cases and 94 controls. Aust NZ J of Psychiatry, 2011. Vol.45(6), 458-465

Clozapine-associated myocarditis: Drug Safety 2007; 30:47-57

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NSQHS Standards Applicable:	☒  Std 1: Clinical Governance ☒  Std 4: Medication Safety		☐  Std 5: Comprehensive Care ☒  Std 8: Recognising and Responding to Acute Deterioration		
National Standards for Mental Health Services	☒ Std 1: Rights and Responsibilities ☐ Std 2: Safety ☐ Std 3: Consumer and Carer Participation ☒ Std 8: Governance, leadership and management		☐ Std 9: Integration ☒ Std 10: Delivery of Care		
Chief Psychiatrist's Standards for Clinical Care:	Assessment, Care Planning, Consumer and Carer Involvement in Individual Care, Physical Health Care of Mental Health Consumers, Risk Assessment and Management, Transfer of Care				
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Document Version Control			
Date	Version	Author	Notes
02/10/2018	1.30	Chief Pharmacist, NMHS Chair DTC	Sponsor approved, minor amendment, updated monitoring for clozapine induced hypomotility and updated cardiac monitoring in accordance with best practice, Change of Sponsor, template updated.
04/22/2019	V1.40	A/Policy and Audit Officer	Addition of review statement
15/05/2020	V2.0	Chief Pharmacist NMHS DTC	Scheduled review, updated monitoring for Clozapine, induced hypomotility
10/10/2022	V3.0	Chief Pharmacist NMHS DTC	Combination of Inpatient & Community Clozapine policies; updated template & links; new information on monitoring.

The health impact upon Aboriginal people have been considered, and where relevant incorporated and appropriately addressed in the development of this health initiative. ISD Number: 1710.

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Appendix 1. Community Commencement Schedule for Prescribing & Monitoring

DAY	DAY OF THE WEEK (SUGGESTED)	AM DOSE (mg)	PM DOSE (mg)	MONITORING
1	Tuesday	6.25	NIL	TPR & postural BP. Monitor prior to giving first dose and then hourly for 3 hours
2	Wednesday	6.25	6.25	TPR & postural BP
3	Thursday	6.25	6.25	TPR & postural BP
4	Friday	6.25	12.5	TPR & postural BP. MSE Weight Actively manage adverse effects**
5	Saturday	12.5	12.5	No routine monitoring unless clinically indicated
6	Sunday	12.5	12.5	No routine monitoring unless clinically indicated
7	Monday	12.5	25	TPR & postural BP. MSE Blood test - FBC, Troponin, CRP
8	Tuesday	12.5	25	TPR & postural BP. MSE & Prescribe
9	Wednesday	25	25	TPR & postural BP.
10	Thursday	25	37.5	TPR & postural BP.
11	Friday	25	37.5	TPR & postural BP, MSE Weight Actively manage adverse effects**
12	Saturday	25	37.5	No routine monitoring unless clinically indicated
13	Sunday	25	37.5	No routine monitoring unless clinically indicated
14	Monday	37.5	37.5	Not seen unless problems Blood test - FBC & CRP, Troponin
15	Tuesday	37.5	37.5	TPR & postural BP. Weight MSE & Prescribe; Actively manage adverse effects**
16	Wednesday	37.5	50	Not seen unless problems
17	Thursday	37.5	50	Not seen unless problems
18	Friday	50	50	TPR & postural BP.
19	Saturday	50	50	No routine monitoring unless clinically indicated
20	Sunday	50	50	No routine monitoring unless clinically indicated
21	Monday	50	75	Not seen unless problems

DAY	DAY OF THE WEEK (SUGGESTED)	AM DOSE (mg)	PM DOSE (mg)	MONITORING
				Blood test - FBC & CRP, Troponin
22	Tuesday	50	75	TPR & postural BP. MSE & Prescribe weight, actively manage adverse effects**
23	Wednesday	75	75	Not seen unless problems
24	Thursday	75	75	Not seen unless problems
25	Friday	75	100	Not seen unless problems
26	Saturday	75	100	No routine monitoring unless clinically indicated
27	Sunday	75	100	No routine monitoring unless clinically indicated
28	Monday	75	100	Not seen unless problems Blood test - FBC, CRP, Troponin
ON-GOING	Weekly review until week 18	As clinically indicated		Refer to Appendix 2 .

**See [Section 10](#) for side effects requiring escalation to medical team

Appendix 2. Ongoing Monitoring >4 weeks following initiation

Ongoing monitoring	Frequency for inpatients	Frequency for community patients	Additional Information
Physical Examination	Annually	Annually	Clozapine monitoring Form Part B
Weight & Waist Circumference	Weekly	Weekly until week 18 and then 4 weekly thereafter	AORC chart and Clozapine monitoring Form Part A Weight chart
TPR & postural BP	Daily	Weekly until week 18 and then 4 weekly thereafter	AORC chart and Clozapine monitoring Form Part A
FBC	Weekly for the first 18 weeks and then every 4 weeks thereafter if blood results within range	Weekly for the first 18 weeks and then every 4 weeks thereafter if blood results within range	Clozapine monitoring Form Part A Form 5 signed by doctor in community. Blood tests to be entered on ClopineCentral® prior to dispensing clozapine
Fasting BGL/HbA1c	3 months after initiation, then repeat at 6 months, and every 6 months ongoing	3 months after initiation, then repeat at 6 months, 12 months and a minimum of annually thereafter.	Clozapine monitoring Form Part B
Lipid profile	3 months after initiation, then at 6 months, and every 6 months ongoing	3 months after starting, then at 6 months, 12 months and a minimum of annually thereafter.	Clozapine monitoring Form Part B
U&Es, LFTs	Repeat every 6 months	6 months after initiation, then at 12 months and a minimum of annually thereafter.	Clozapine monitoring Form Part B
Troponin/CRP	Repeat if clinically indicated	Repeat if clinically indicated	Clozapine monitoring Form Part B
ECG	Repeat every 6 months	Repeat when target dose is reached and then a minimum of annually thereafter.	Additional ECGs should be performed as clinically indicated especially for patients on other medications that prolong the QTc. Clozapine monitoring Form B.

Echocardiogram	Repeat at 12-months, 2 years, and 5 years - If NO abnormality or concern, repeat every 5 years ongoing. - If YES to abnormality (even mild) continue annually	Repeat at 12-months, 2 years, and 5 years - If NO abnormality or concern, repeat every 5 years ongoing. - If YES to abnormality (even mild) continue annually	Repeat at any time if clinically indicated. Clozapine monitoring Form Part B
Clozapine Levels	Check clozapine level once dose is stabilised and then as clinically indicated. 6-monthly checking of clozapine level with routine bloods tests is recommended.	Check clozapine level once dose is stabilised and then as clinically indicated. Annual checking of clozapine level with routine bloods tests is recommended.	Clozapine levels may help with optimising dosing, monitoring of adherence, drug interactions, the effects of smoking changes and troublesome adverse effects. If patient has inadequate response to clozapine, then trough levels should be optimised between 350-600mcg/L. Trough levels > 600mcg/L are associated with an increased risk of SEs. Repeat levels after 5-7 days following dose changes or changes in smoking status. Clozapine monitoring Form Part B
Constipation	Ask about bowel habits and document on bowel chart a minimum of weekly.	Ask about bowel habits a minimum of weekly for the first 18 weeks and then a minimum of 4 weekly thereafter.	Use Rome IV criteria. Low threshold for starting laxatives. Prophylactic laxatives may be required. Clozapine monitoring Form Part A Bowel chart
Smoking Status		Ask about smoking status and document at every visit.	Clozapine monitoring Form Part A

Note:

- Frequency of monitoring should be increased if clinically indicated
- Most routine blood tests
 - Minimum of every 6 months for inpatients
 - minimum of every 12 months for community patients after the first year
- Weight gain with clozapine is usually most significant in the first year so more frequent monitoring of glucose and lipids are recommended in this time. Note - Dyslipidaemia and hyperglycaemia may occur with or without weight gain.